



#### △ Lilienthal "Normal Apparatus" 1894

**Origin** Germany

**Engine** None

**Top speed** N/A

Designed by Otto Lilienthal, the *Normalsegelapparat* or "Normal soaring apparatus" is considered to be the first flying machine to enter serial production; at least nine were built.

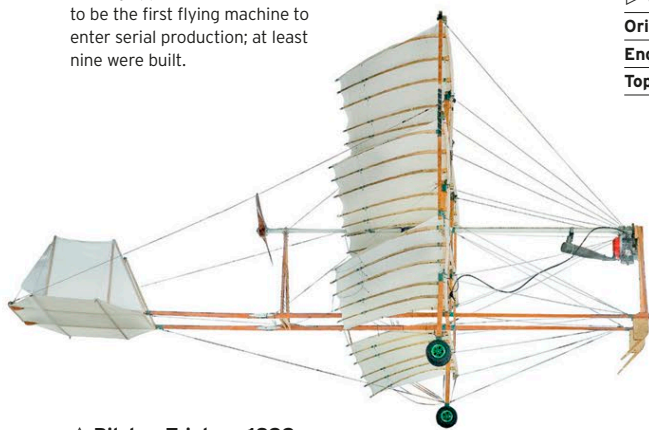
#### ▷ Chanute biplane 1896

**Origin** USA

**Engine** None

**Top speed** N/A

Retired civil engineer Octave Chanute devoted himself to advancing the new science of aviation. Although he did not invent the biplane, the "strut-braced" design is credited to him.



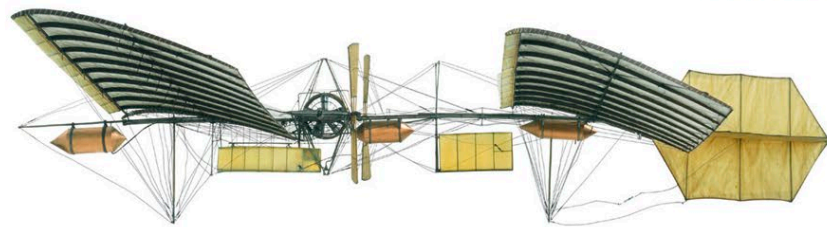
#### △ Pilcher Triplane 1899

**Origin** UK

**Engine** 4 hp air-cooled twin

**Top speed** 26 mph (42 km/h)

Pilcher's correspondence with Octave Chanute had convinced him that the way to generate adequate lift without an unfeasibly large wingspan was to stack wings in layers—hence the triplane. Sadly, he died before his powered triplane flew. This replica has flown.



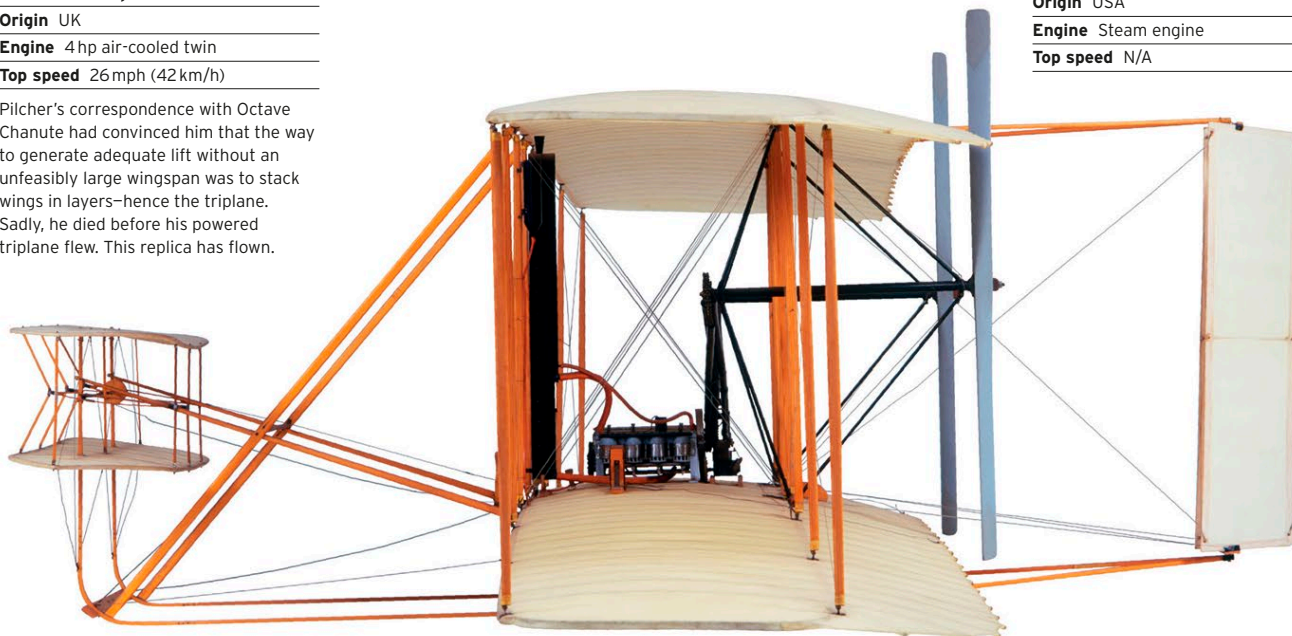
#### △ Langley Aerodrome 1903

**Origin** USA

**Engine** Steam engine

**Top speed** N/A

While the steam-driven scale models flew well, Langley's full-size Aerodrome did not. Launched by catapult from a houseboat on the Potomac River, each attempt ended in disaster.



#### ◁ Wright Flyer 1903

**Origin** USA

**Engine** 12 hp Taylor liquid-cooled in-line 4

**Top speed** N/A

Undoubtedly one of the most important machines of all time, the Flyer demonstrated unequivocally that powered, controlled flight was possible. It made four flights on December 17, 1903, before being blown over and badly damaged.